MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, TYPES 1N3289, 1N3291, 1N3293, 1N3294, 1N3295, AND R TYPES, JAN, JANTX, JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon power rectifiers. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Normal and reverse types (reverse types, suffix R). Reverse and normal types are identical except the normal types have the cathode connected to the stud and the reverse types have the anode connected to the stud. Designated values are applicable to both types.

1.3 Physical dimensions. See figure 1 (DO-205AA - formerly DO-8).

1.4 Ratings and characteristics.

Type	V_{RSM}	V_{RWM}	I_O T_C = 150°C	I_O [1/] T_C = 134°C	I_{FSM} 1/120s	Barometric pressure (reduced)	T_J and T_{STG}
	V (pk)	V (pk)	A dc	A dc	A (pk)	mm Hg	°C
1N3289	240	200	75	100	1600	---	-65
1N3291	480	400	75	100	1600	8	to
1N3293	720	600	75	100	1600	8	
1N3294	960	800	75	100	1600	15	+200
1N3295	1200	1000	75	100	1600	15	

1/ Derate linearly 1.5 A dc/°C for $T_C \geq 134°C$.

1.5 Thermal resistance characteristic: $R_{\theta JC}$ = 0.4°C/W.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Electronics Research and Development Command, ATTN: DELET-R-S, Fort Monmouth, NJ 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specification and standard, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figure 1 herein.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

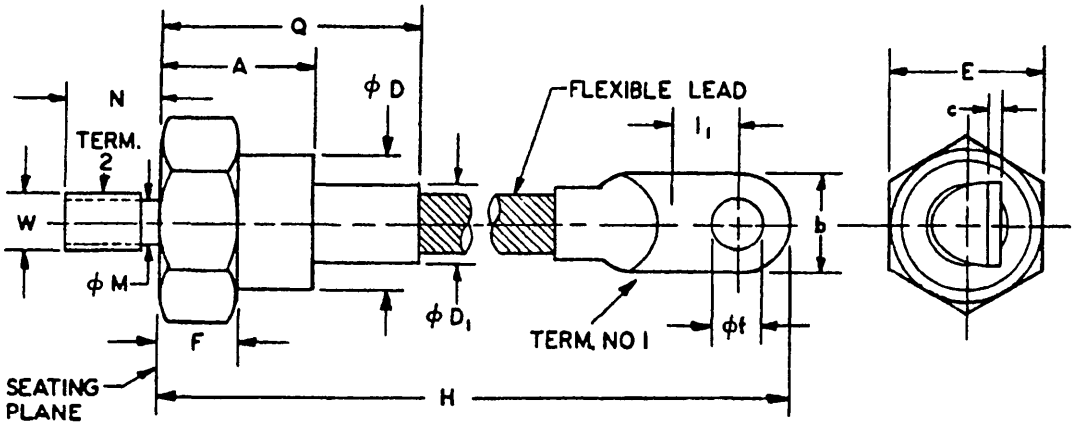
4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein.

Measurements required by the applicable screening steps of MIL-S-19500 are as follows, with the limits in accordance with table I herein.

Screen (see table II of MIL-STD-19500)	Measurement
	JANTX and JANTXV levels
1	Method 2073 may be used in lieu of 2074 for compression bonded devices only.
4	5000G; Performed prior to installation of external lead.
10	Not applicable.
11	I_{RM1}, and V_{FM} of subgroup 2 of table I herein.
12	Method 1038, condition A; $V_{RM} = V_{RWM}$ rated (see 1.4), $T_C = 150^{\circ}C$; f = 60 Hz.
13	Subgroup 2 of table I herein: $\Delta I_{RM1} = \pm 0.5$ mA (pk) or 100% of initial value whichever is greater. $\Delta V_{FM} = \pm 0.1$ V (pk).

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SYMBOL	Dimensions				NOTES
	Inches		Millimeters		
	Min	Max	Min	Max	
A		1.140		28.96	
b		.670		17.02	
c	.050	.120	1.27	3.05	
ØD	.625	1.000	15.88	25.40	
$ØD_1$		.500		12.70	
E	1.031	1.063	26.19	27.00	
F	.125	.500	3.18	12.70	5
H	4.300	5.065	109.22	128.65	
I_1	.300	.450	7.62	11.43	6
ØM	.343	.373	8.71	9.47	3
N	.605	.645	15.37	16.38	
Q		1.750		44.45	
Øt	.250	.310	6.35	7.87	
W					4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Complete threads to extend to within 2-1/2 threads of seating plane.
4. .375-24 UNF-2A. Maximum pitch diameter of plated threads shall be basic pitch diameter (.3479 (8.837 mm) inches) reference.
5. A chamfer or undercut on one or both ends of hexagonal portions is optional.
6. Minimum flat.
7. For marking see 3.4.

FIGURE 1. Physical dimensions.

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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. Electrical measurements shall be in accordance with the applicable steps of table IV herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb of MIL-S-19500, and table II herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end points) shall be in accordance with the applicable steps of table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Power cycling. One complete test shall be as follows: First, heat the case to the maximum temperature specified by passing the specified amount of forward current through the diode under test. Then, remove the applied current and allow the case temperature to cool to the minimum case temperature specified. No time limit is applicable to any one cycle, but the cycling shall be continuous until the required number of cycles have been completed.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, OS, SH

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5961-0946)

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TABLE I. Group A inspection.

Inspection	MIL-STD-750		LTPD 1/	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 1							
Visual and mechanical inspection	2071						
Subgroup 2							
Forward voltage	4011	Pulse method: I_{FM} = 310 A (pk); pulse width = 8.3 ms max; duty cycle = 2% max		V_{FM1}	---	1.55	V (pk)
Reverse current	4016	AC method:		I_{RM1}			
1N3289		V_{RM} = 200 V (pk)			---	10	mA (pk)
1N3291		V_{RM} = 400 V (pk)			---	10	mA (pk)
1N3293		V_{RM} = 600 V (pk)			---	10	mA (pk)
1N3294		V_{RM} = 800 V (pk)			---	10	mA (pk)
1N3295		V_{RM} = 1000 V (pk)			---	10	mA (pk)
Reverse current at peak reverse voltage	4016	AC method:		I_{RM2}			
1N3289		V_{RM} = 240 V (pk)			---	40	mA (pk)
1N3291		V_{RM} = 480 V (pk)			---	40	mA (pk)
1N3293		V_{RM} = 720 V (pk)			---	40	mA (pk)
1N3294		V_{RM} = 960 V (pk)			---	40	mA (pk)
1N3295		V_{RM} = 1200 V (pk)			---	40	mA (pk)
Subgroup 3							
High temperature operation:		T_C = 200°C					
Reverse current	4016	AC method:		I_{RM3}			
1N3289		V_{RM} = 200 V (pk)			---	30	mA (pk)
1N3291		V_{RM} = 400 V (pk)			---	30	mA (pk)
1N3293		V_{RM} = 600 V (pk)			---	30	mA (pk)
1N3294		V_{RM} = 800 V (pk)			---	30	mA (pk)
1N3295		V_{RM} = 1000 V (pk)			---	30	mA (pk)
Low-temperature operation:		T_C = -65°C					
Forward voltage	4011	Pulse method: I_{FM} = 310 A (pk); pulse width = 8.3 ms max; duty cycle = 2% max		V_{FM2}	---	1.45	V (pk)
Reverse current	4016	AC method:		I_{RM4}			
1N3289		V_{RM} = 200 V (pk)			---	40	mA (pk)
1N3291		V_{RM} = 400 V (pk)			---	40	mA (pk)
1N3293		V_{RM} = 600 V (pk)			---	40	mA (pk)
1N3294		V_{RM} = 800 V (pk)			---	40	mA (pk)
1N3295		V_{RM} = 1000 V (pk)			---	40	mA (pk)

See footnotes at end of table.

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TABLE I. Group A inspection - continued.

Inspection	MIL-STD-750		LTPD 1/	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 4							
Not applicable							
Subgroup 5							
Not applicable							
Subgroup 6							
Surge current	4066	T_C = 150°C; I_{FSM} = 1600 A (pk); I_O = 75 A dc; ten 1/120 s surges; 1 surge/minute V_{RM} = V_{RWM} (see 1.4)					
Electrical measurements		See table IV, steps 1 and 2					
Subgroup 7							
Not applicable							

1/ For LTPD values, see MIL-S-19500.

TABLE II. Group B inspection.

Inspection	MIL-STD-750		LTPD 1/
	Method	Conditions	
Subgroup 1			
Resistance to solvents	1022		
Subgroup 2			
Thermal shock (temperature cycling)	1051		
Hermetic seal	1071		
a. Fine leak			
b. Gross leak		Test condition C, step 2 or test condition D	
Electrical measurements		See table IV, steps 1 and 2	

See footnotes at end of table.

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TABLE II. Group B inspection - continued.

Inspection	MIL-STD-750		LTPD
	Method	Conditions	
Subgroup 3			
Blocking life	---	T_C = 200°C; V_{RWM} = V_{RM} (see 1.4); I_O = 0; half-sine wave, 60 Hz; t = 340 hours	
Electrical measurements		See table IV, steps 1 and 2 (step 2 should be performed before step 1)	
Subgroup 4			
Decap internal visual design verification	2075		
Subgroup 5			
Thermal resistance	3151	$R_{\theta JC}$ = 0.4°C/W	
Subgroup 6			
High temperature (nonoperating) life	1032	T_A = 200°C	
Electrical measurements		See table IV, steps 1 and 2	
Subgroup 7			
Power cycling	---	See 4.5.1, I_O = 75 A dc; T_C (low) = 40°C; T_C (high) = 150°C; 5000 cycles	
Electrical measurements		See table IV, steps 1, 2, and 3	

1/ For LTPD values, see MIL-S-19500.

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TABLE III. Group C inspection.

Inspection	MIL-STD-750		LTPD 1/
	Method	Conditions	
Subgroup 1			
Physical dimensions	2066	See figure 1	
Subgroup 2			
Thermal shock (glass strain)	1056	Test condition B	
Terminal strength	2036		
Tension (lead)		Test condition A; weight = 100 lb; t = 15 ±3 s	
Torque (terminal)		Test condition D1, 10 lb-in; t = 15 ±3 s	
Torque (stud)		Test condition D2; 125 lb-in; t = 15 ±3 s	
Hermetic seal	1071		
a. Fine leak			
b. Gross leak		Test condition C, step 2, or test condition D	
Moisture resistance	1021		
Visual and mechanical inspection	2071		
Electrical measurements		See table IV, steps 1 and 2	
Subgroup 3			
Shock	2016	500 G; t = 1 ms; 5 blows each orientation: X1, Y1, and Y2	
Vibration, variable frequency	2056	10 G, nonoperating	
Constant acceleration	2006	5,000 G; X1, Y1, and Y2 orientations	
Electrical measurements		See table IV, steps 1 and 2	

See footnotes at end of table.

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TABLE III. Group C inspection - continued.

Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	LTPD 1/
Subgroup 4			
Salt atmosphere (corrosion)	1046	Test condition B	
Subgroup 5			
Barometric pressure, reduced (altitude operation)	1001	t = 60 s	
1N3291, 1N3293		Pressure = 8 mm Hg	
1N3294, 1N3295		Pressure = 15 mm Hg	
Measurement during test		See table IV, step 2	
Subgroup 6			
Blocking life	---	T_C = 200°C; V_{RWM} = V_{RM} (see 1.4); I_O = 0; half-sine wave; f = 60 Hz; t = 1000 hrs	
Electrical measurements		See table IV, steps 1 and 2 (step 2 shall be performed before step 1)	

1/ For LTPD values, see MIL-S-19500.

TABLE IV. Groups A, B, and C end-point electrical measurements.

Step	Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
1.	Forward voltage	4011	Pulse method: I_{FM} = 310 A (pk); pulse width = 8.3 ms max; duty cycle = 2% max	V_{FM}	---	1.55	V (pk)
2.	Reverse current	4016	AC method:	I_{RM1}			
	1N3289		V_{RM} = 200 V (pk)		---	10	mA (pk)
	1N3291		V_{RM} = 400 V (pk)		---	10	mA (pk)
	1N3293		V_{RM} = 600 V (pk)		---	10	mA (pk)
	1N3294		V_{RM} = 800 V (pk)		---	10	mA (pk)
	1N3295		V_{RM} = 1000 V (pk)		---	10	mA (pk)
3.	Forward voltage (change)	4011	I_{FM} = 310 A (pk)	ΔV_{FM}	---	+0.1	V (pk)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-19500/246F	Semiconductor Device, Diode, Silicon, Power Rectifier, Types 1N3289, 1N3291, 1N3293, 1N3295, and R Types.

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
JAN, JANTX, JANTXV	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
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5. PROBLEM AREAS

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